



Inspiring Excellence

CSE370 : Database Systems

Project Report

Project Title : SCMS - Student Course Management System

Group No : 03, CSE370 Lab Section : 10, Spring 2026		
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Introduction

Our project is a **Student Course Management System (SCMS)**. It is a web application that helps students plan their courses, avoid schedule conflicts, calculate CGPA, track academic tasks, find available rooms, share course resources, and book consultations with faculty. Faculty members can manage their advisee's, consultation slots, and course resources. The main goal of SCMS is to make academic planning easier for students while allowing faculty to manage their tasks efficiently.

Project Features

1. Intelligent Course Prerequisite Planner

Students input their current semester and completed courses. The system validates prerequisite requirements and generates a curated list of eligible courses for the upcoming semester, ensuring students pursue courses they are academically qualified to take.

2. Conflict-Free Schedule Generator

When students select specific course sections, the system cross-references class timings and notifies them of any scheduling conflicts. This prevents overlaps and ensures students can attend all selected courses without conflicts.

3. Dynamic CGPA & Target Grade Predictor

A sophisticated calculator that tracks historical academic performance. Students can set a target CGPA, and the system computes the exact grades required in current courses to achieve that goal, providing clarity on academic objectives and required effort.

4. Interactive Course Planner

Students select courses for the upcoming semester through an intuitive interface. The system calculates total credit hours and enforces credit-load limitations to prevent registration overload and maintain academic sustainability.

5. Academic Task Tracker

A Kanban-style task management board for organizing assignments and exams. The system automatically links tasks to enrolled courses and sorts them by due date, keeping students focused on immediate priorities.

6. Room & Lab Availability Finder

Students can search for unoccupied study spaces in real-time. The system queries the schedule database against the current time or a specified time block, delivering availability information suitable for group study or independent work.

7. Course Resource Hub

A centralized repository for student-shared academic resources, including notes, video tutorials, and study guides organized by course. An upvote/downvote mechanism ensures highly-rated resources appear at the top.

8. Faculty Consultation Booking

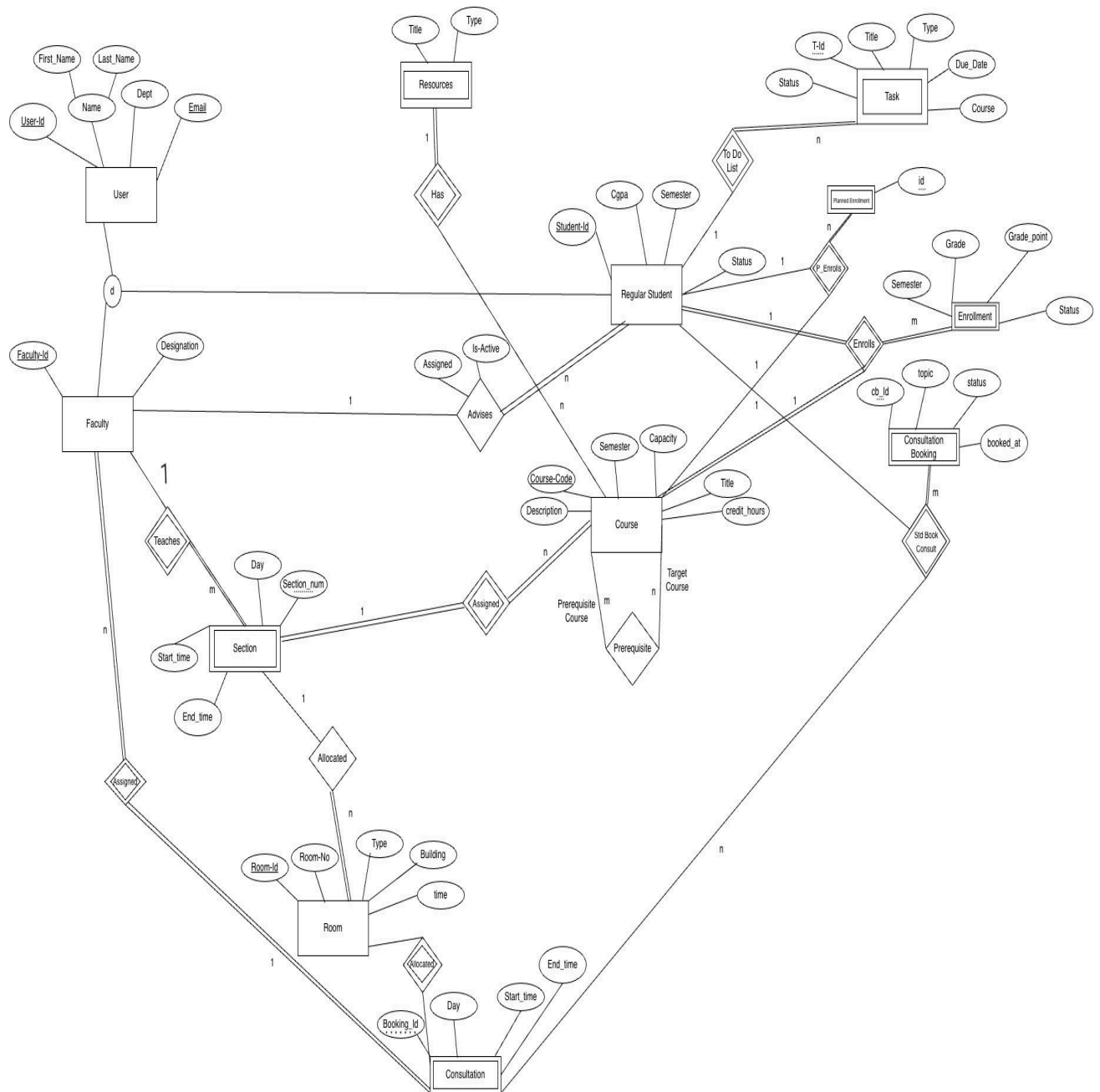
A directory displaying faculty office hours with an integrated booking system. Students can reserve consultation slots using a calendar interface while preventing double-booking conflicts.

9. Faculty & Advisor Portal

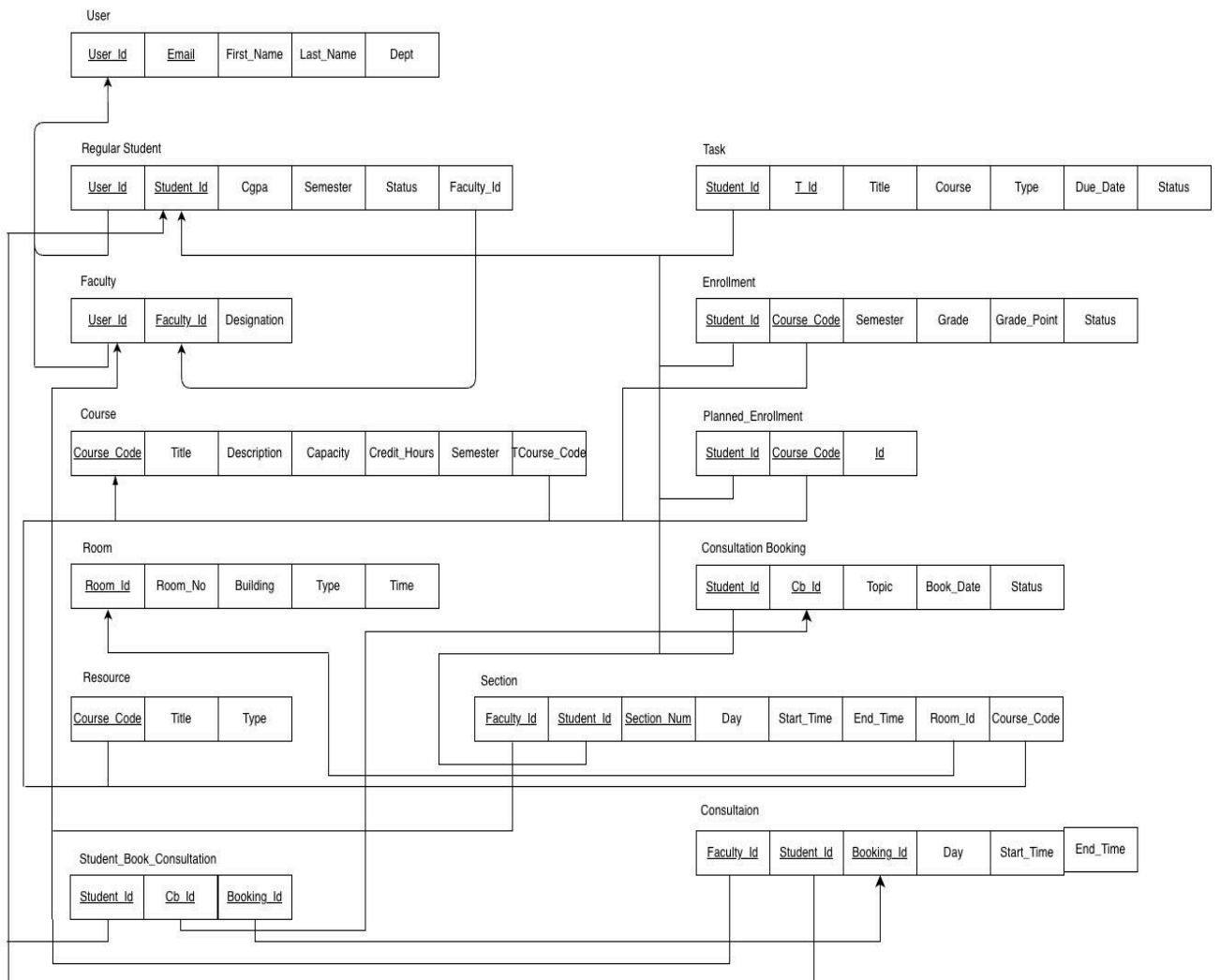
Faculty members can log in to confirm or deny consultation requests and serve as academic advisors, guiding students in course selection and academic planning.

ID, Name	Features [3 per member]	
23201387, Sajir Hamid	Ft 1	Intelligent Course Prerequisite Planner
	Ft 2	Conflict-Free Schedule Generator
	Ft 3	Dynamic CGPA & Target Grade Predictor
23201589, Afifa Tayeaba	Ft 1	Interactive Course Planner
	Ft 2	Academic Task Tracker
	Ft 3	Faculty Consultation Booking
22201382, Md Nafiul Hasan	Ft 1	Course Resource Hub
	Ft 2	Room & Lab Availability Finder
	Ft 3	N/A

ER/EER Diagram



Schema Diagram



Normalization

- a. Explain if your converted Schema is in 1NF or not. If not, decompose it to 1NF.

Ans: Yes, It is in 1NF form. Because there is not multivalued, composite and nested attribute in the schema table.

- b. Explain if your converted Schema is in 2NF or not. If not, decompose it to 2NF. Can there be any partial functional dependencies in your relational schema?

Ans: Yes, It is in 2NF form. Because, It is in 1NF form & no functional dependencies appear.

- c. Explain if your converted Schema is in 3NF or not. If not, decompose it to 3NF. Can there be any transitive dependencies in your relational schema?

Ans: Yes, It is in 3NF form. Because It is in 2NF form & no transitive dependencies appear.

Frontend Development

The frontend of our project was built using **React**, **Vite**, **Tailwind CSS**, and **shadcn/ui components**. The user interface is responsive and interactive. Students can view eligible courses, track CGPA, manage tasks, check room availability, and book consultations. Faculty members can view advisee's, manage consultations, and post course resources.

Contribution of ID : 23201387, Name : Sajir Hamid
Everyone contributed

Contribution of ID : 23201589, Name : Afifa Tayeaba
Everyone contributed

Contribution of ID : 22201382, Name : Md Nafiul Hasan
Everyone contributed

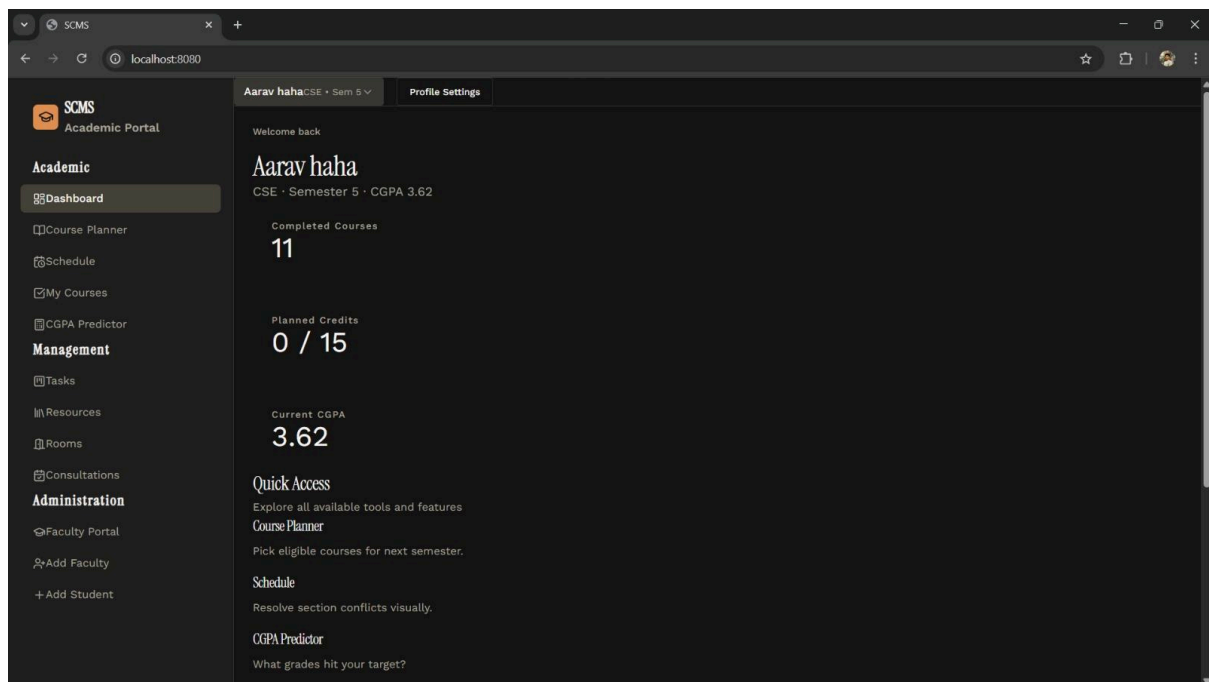


Fig- Dashboard of SCMS

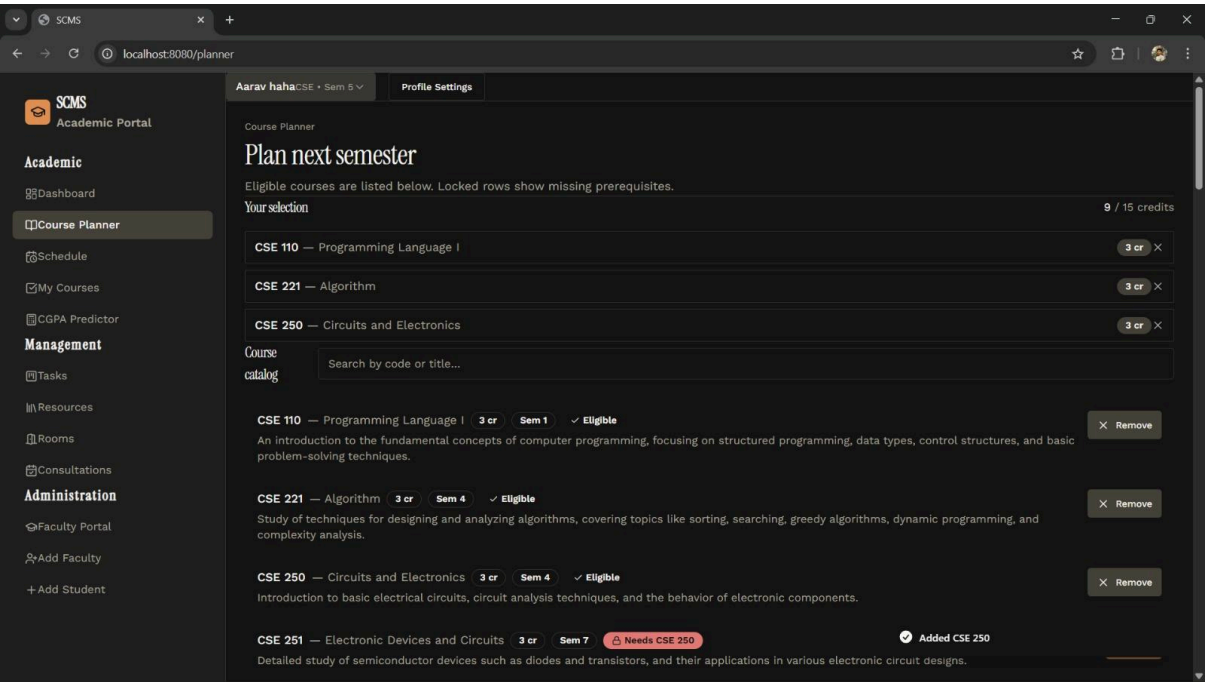


Fig- Course Planner Feature

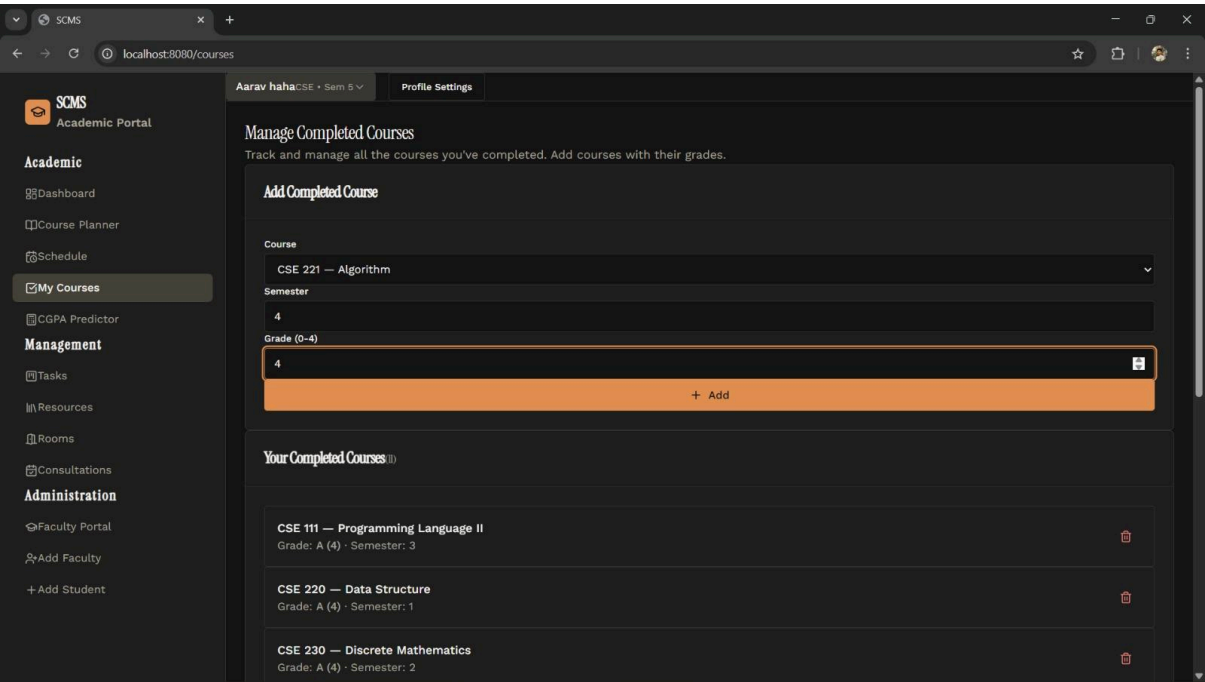


Fig- My Courses Shows My Completed Course List

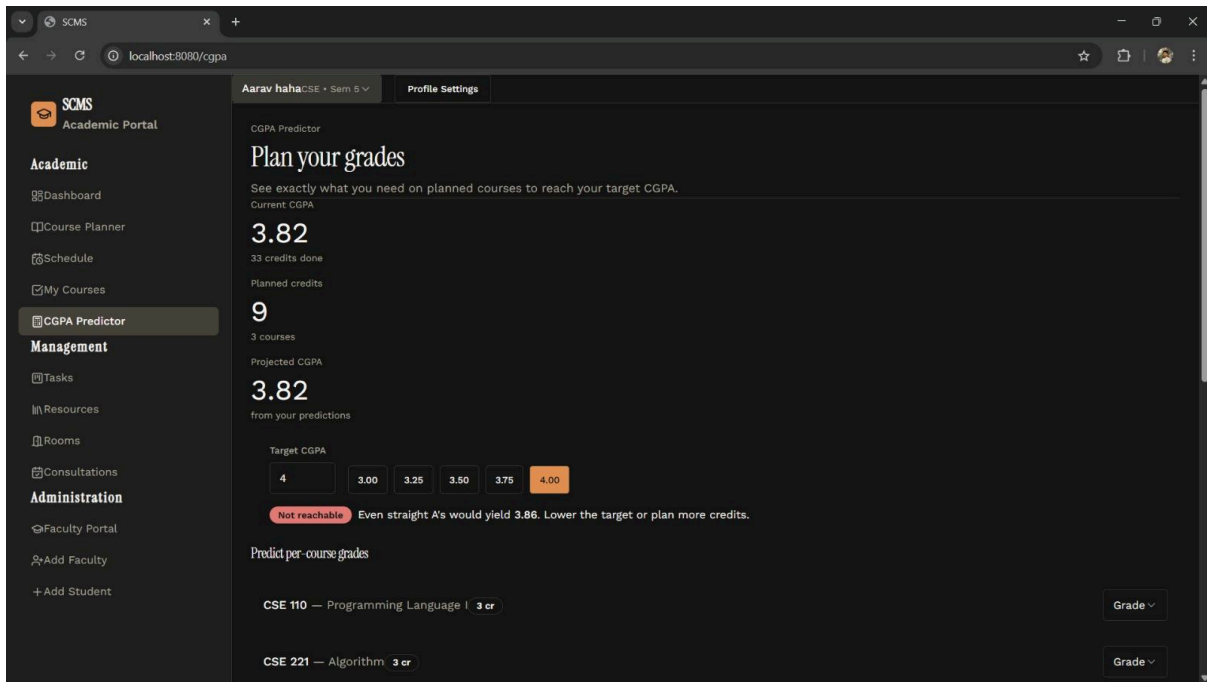


Fig- CGPA Predictor Based On Target

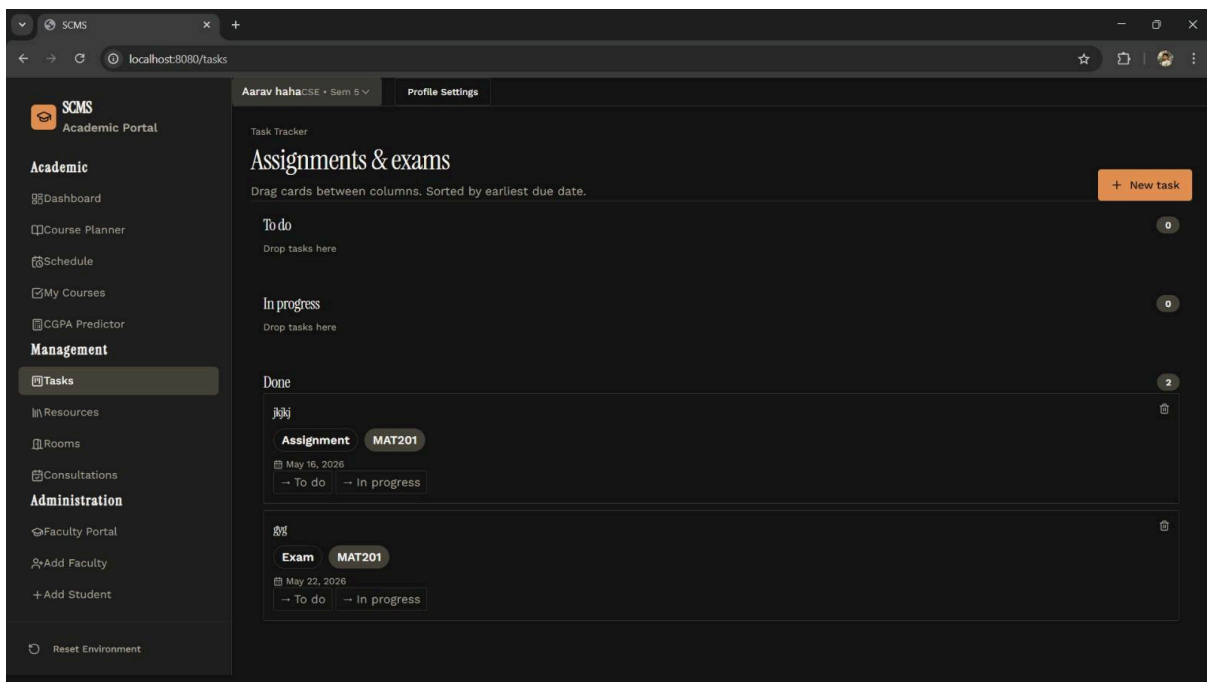


Fig- Task Management Feature

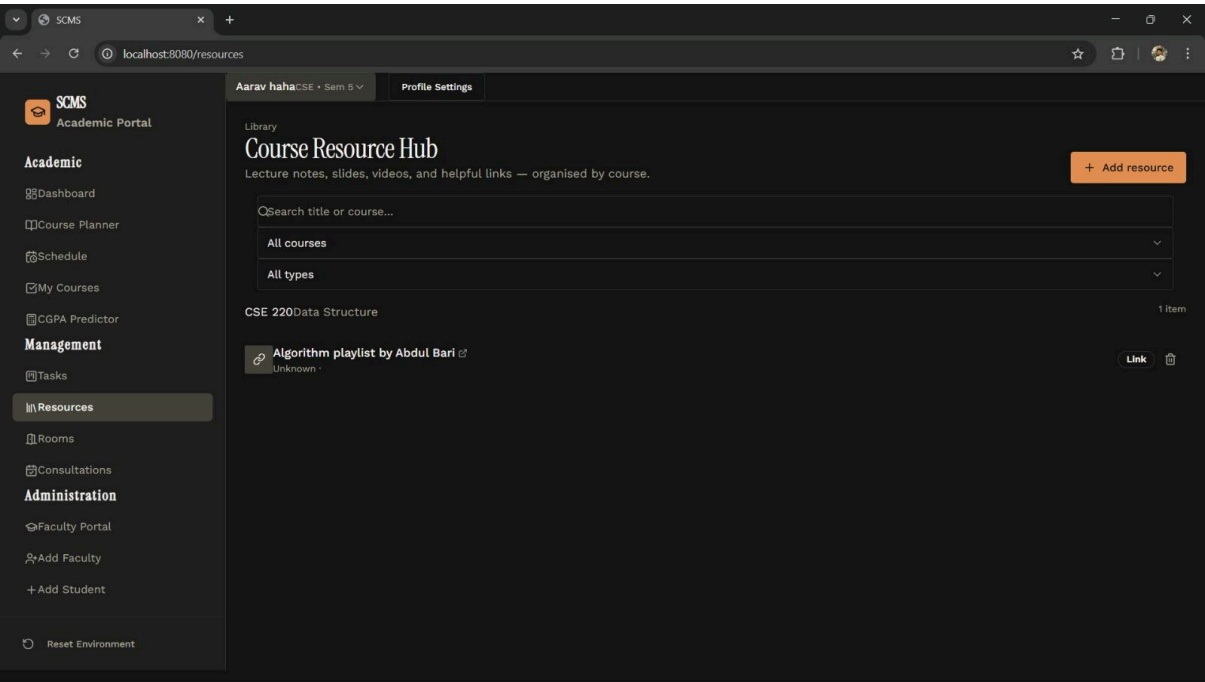


Fig- Resource Hub Management Feature

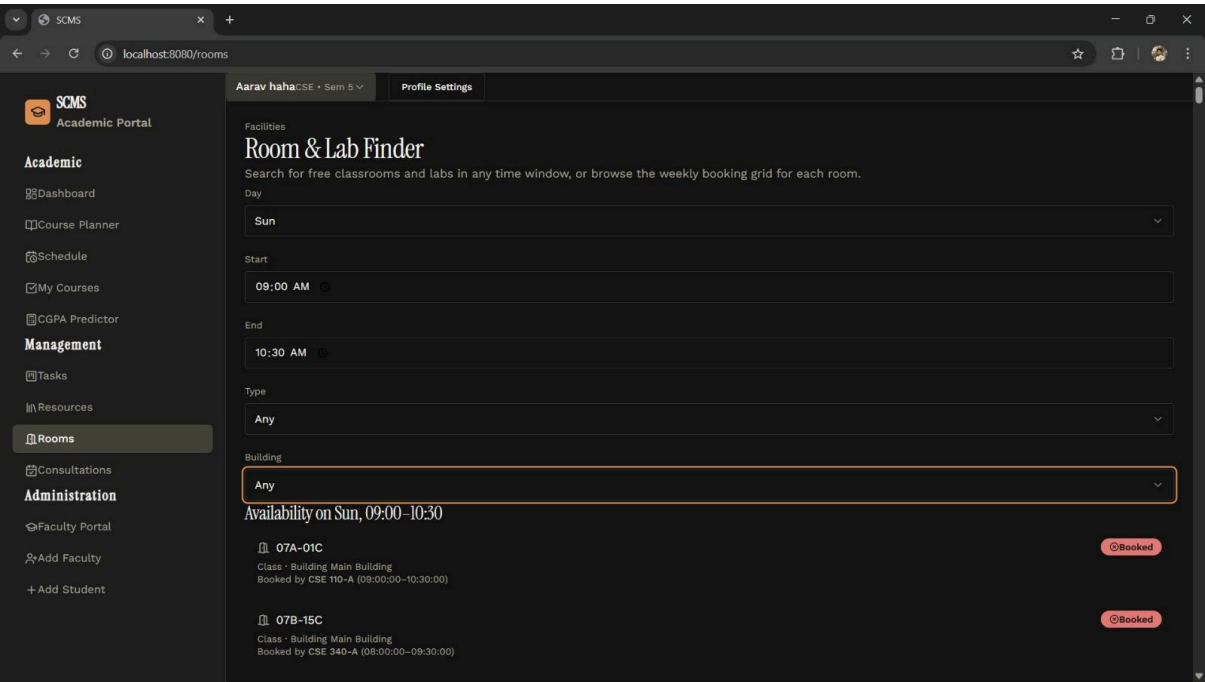


Fig- Room & Lab Finder Feature

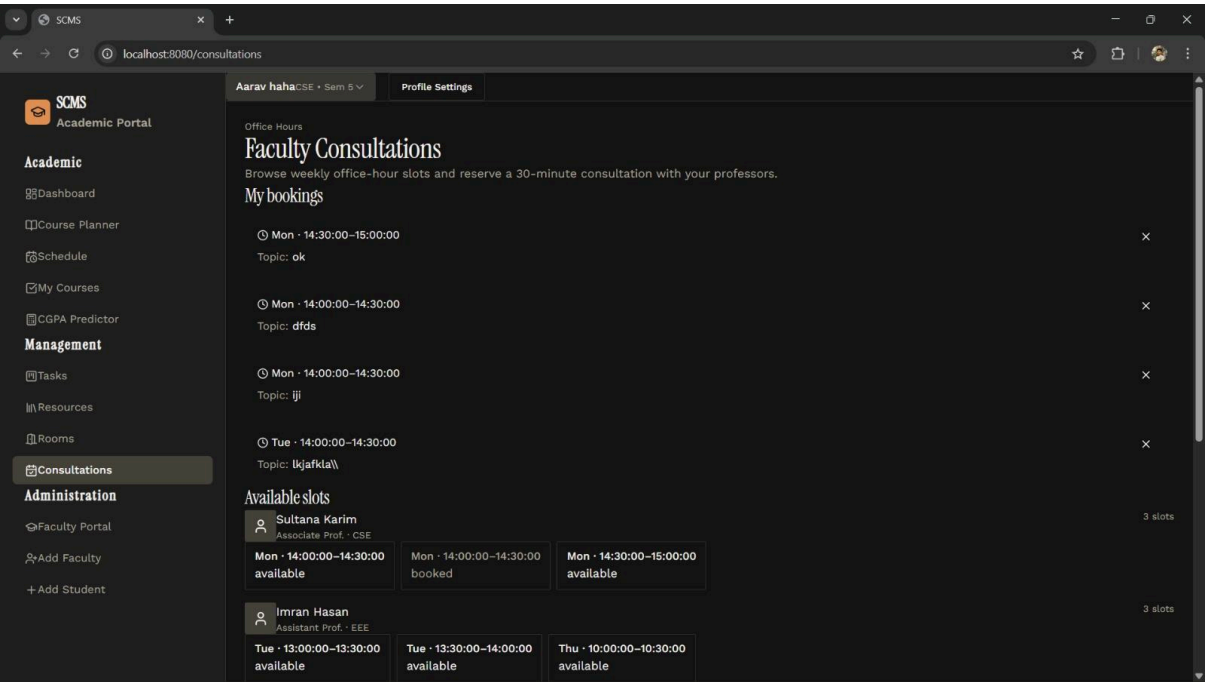


Fig- Consultation Feature

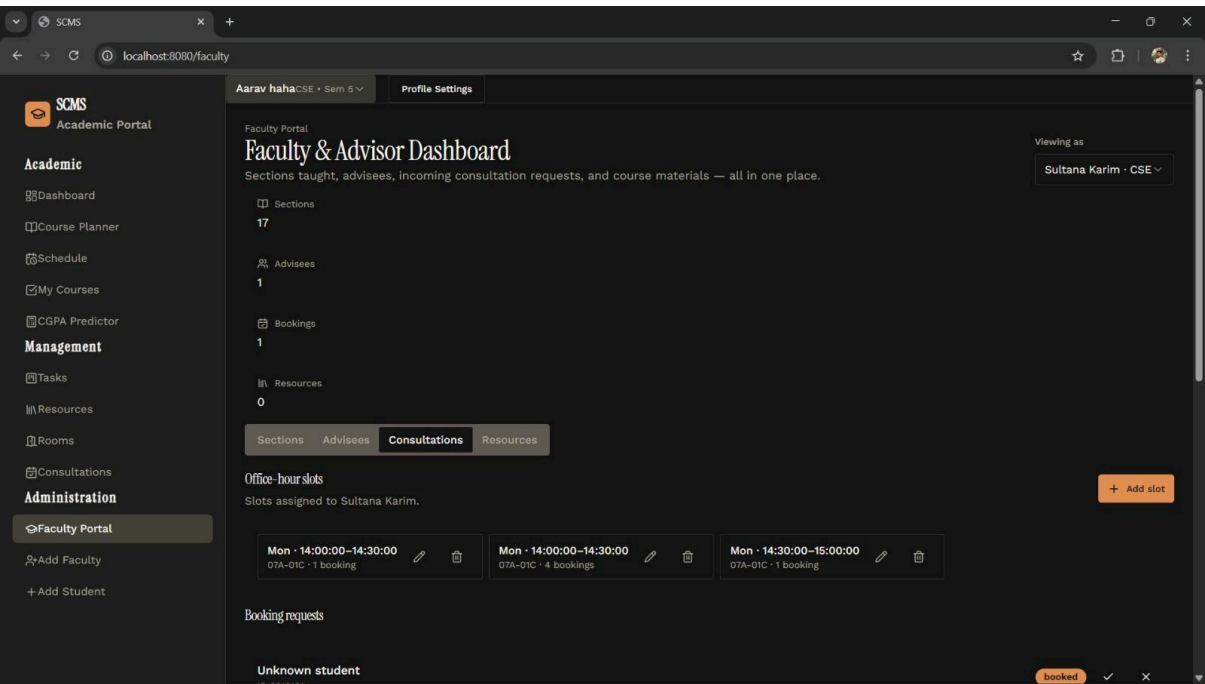


Fig- Faculty Panel Layout

The screenshot shows a web browser window with the URL `localhost:8080/faculty/new`. The page is titled "Create Faculty Account" and is part of the SCMS Academic Portal. The left sidebar contains navigation links under "Academic" (Dashboard, Course Planner, Schedule, My Courses, CGPA Predictor) and "Management" (Tasks, Resources, Rooms, Consultations). Under "Administration", the "Faculty Portal" is selected, and the "Add Faculty" button is highlighted. The main content area has a header "Aarav hahacse • Sem 5" and a "Profile Settings" tab. The form is titled "Faculty Details" and includes input fields for "First name", "Last name", "Email", and "Department". There is a dropdown menu for "Designation" with the option "Select designation". A "Create Faculty" button is at the bottom of the form. The bottom of the browser window shows a Windows taskbar with a search bar and various application icons, including a weather widget showing 27°C and rain showers.

Fig- Add Faculty Layout

The screenshot shows a web browser window with the URL `localhost:8080/students/new`. The page is titled "Create Student" and is part of the SCMS Academic Portal. The left sidebar is identical to the previous screenshot, but the "+ Add Student" button under the "Administration" section is highlighted. The main content area has a header "Aarav hahacse • Sem 5" and a "Profile Settings" tab. The form is titled "Create Student" with a subtitle "Fill basic details, then add completed courses." It includes input fields for "First name", "Last name", "Student ID", "Email", "Department", "Semester", and "CGPA (optional)". At the bottom of the form are "Create Student" and "Cancel" buttons. The bottom of the browser window shows a Windows taskbar with a search bar and various application icons, including a weather widget showing 27°C and rain showers.

Fig- Add Student Layout

Backend Development

The backend is built using **plain PHP API files** located in src/apis and communicates with a **MySQL/MariaDB database** named scms. Each PHP file represents an API endpoint that handles requests like fetching courses, enrolling students, managing tasks, and booking consultations. The React frontend calls these APIs using fetch.

Key backend functionalities include:

1. **Course Prerequisite Checker:** Validates if a student can enroll in a course based on completed prerequisites.
2. **Conflict-Free Schedule Generator:** Ensures that course sections do not overlap.
3. **CGPA and Target Grade Predictor:** Calculates current CGPA and predicts grades needed to achieve a target CGPA.
4. **Task Tracker Management:** Manages academic tasks linked to courses.
5. **Room and Lab Finder:** Shows free rooms and labs based on student inputs.
6. **Resource Hub:** Manages course materials and rankings based on upvotes and downvotes.
7. **Faculty Consultation Booking:** Allows students to book consultation slots with faculty members.

Contribution of ID : 23201387, Name : Sajir Hamid

Intelligent Course Prerequisite Planner

Conflict-Free Schedule Generator

Dynamic CGPA & Target Grade Predictor

Faculty & Advisor Portal

Contribution of ID : 23201589, Name : Afifa Tayeaba

Interactive Course Planner

Academic Task Tracker

Faculty Consultation Booking

Contribution of ID : 22201382, Name : Md Nafiul Hasan

Course Resource Hub

Room & Lab Availability Finder

```
src > apis > config.php
sajirhtmi, 18 hours ago | 1 author (sajirhtmi)
1 <?php
2 header("Access-Control-Allow-Origin: *");
3 header("Access-Control-Allow-Methods: GET, POST, PUT, DELETE, OPTIONS");
4 header("Access-Control-Allow-Headers: Content-Type");
5 header("Content-Type: application/json; charset=UTF-8");
6
7 if ($_SERVER['REQUEST_METHOD'] === 'OPTIONS') { http_response_code(200); exit; }
8
9 $conn = new mysqli("localhost", "root", "", "scms");
10 if ($conn->connect_error) {
11     http_response_code(500);
12     echo json_encode(["error" => "DB connection failed: " . $conn->connect_error]);
13     exit;
14 }
15 $conn->set_charset("utf8");
16
```

Fig- Portion of Backend Code

```
src > apis > tasks.php
sajirhtmi, 18 hours ago | 1 author (sajirhtmi)
1 require_once __DIR__ . '/config.php';
2 $method = $_SERVER['REQUEST_METHOD'];
3
4 if ($method === 'GET') {
5     $ssid = isset($_GET['student_id']) ? intval($_GET['student_id']) : null;
6     $sql = "SELECT t_id, Student_id AS student_id, Title AS title, Course AS course, Status AS status, Type AS type
7         FROM task WHERE Student_id = ?";
8     if ($ssid) $sql .= " WHERE Student_id = ?";
9     $res = $conn->query($sql);
10    $rows = [];
11    while ($r = $res->fetch_assoc()) $rows[] = $r;
12    echo json_encode($rows);
13 } elseif ($method === 'POST') {
14     $data = json_decode(file_get_contents("php://input"), true);
15     $stmt = $conn->prepare("INSERT INTO task (Student_id, Title, Course, Status, Type, Due_Date) VALUES (?, ?, ?, ?, ?, ?)");
16     $stmt->bind_param("ssssss", $data['student_id'], $data['title'], $data['course'], $data['status'], $data['type'], $data['due_date']);
17     if ($stmt->execute()) {
18         echo json_encode(["success" => true, "t_id" => $conn->insert_id]);
19     } else {
20         http_response_code(400);
21         echo json_encode(["error" => $stmt->error]);
22     }
23 } elseif ($method === 'PUT') {
24     $data = json_decode(file_get_contents("php://input"), true);
25     $existing = null;
26     if (empty($data['t_id'])) {
27         $lookup = $conn->prepare("SELECT Title, Course, Status, Type, Due_Date FROM task WHERE t_id = ?");
28         $lookup->bind_param("i", $data['t_id']);
29         $lookup->execute();
30         $existing = $lookup->get_result()->fetch_assoc() ? null;
31     }
32     $title = $data['title'] ?? ($existing['Title'] ?? null);
33     $course = $data['course'] ?? ($existing['Course'] ?? null);
34     $status = $data['status'] ?? ($existing['Status'] ?? null);
35     $type = $data['type'] ?? ($existing['Type'] ?? null);
36     $duedate = $data['due_date'] ?? ($existing['Due_Date'] ?? null);
37
38     $stmt = $conn->prepare("UPDATE task SET Title=?, Course=?, Status=?, Type=?, Due_Date=? WHERE t_id=?");
39     $stmt->bind_param("sssssi", $title, $course, $status, $type, $duedate, $data['t_id']);
40     $stmt->execute();
41     echo json_encode(["success" => true]);
42 } elseif ($method === 'DELETE') {
43     $data = json_decode(file_get_contents("php://input"), true);
44     $stmt = $conn->prepare("DELETE FROM task WHERE t_id = ?");
45     $stmt->bind_param("i", $data['t_id']);
46     $stmt->execute();
47     echo json_encode(["success" => true]);
48 }
49
50
```

Fig- Portion of Backend Code

